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Teleparallel/$f(T)$

Many models yield $c_T = 1$ at LO; torsion irreps and matter couplings are model-dependent, and our route-equality/flux-ratio diagnostics (C2) do not directly apply. We therefore treat teleparallel cases as outside the present "three-route" posture.

EC/ECSK-like

With propagating torsion or nonminimal fermion couplings, axial/traceless torsion irreps can be present and Dirac axial contacts typically survive. Our C1 pure-trace map excludes these at the analyzed order; see classic EC/MAG treatments [4–6,8].

NLO Dispersion and Spurion Tests

The $\delta c_T^2 \propto k^2$ slope follows from the projected, $PT$-even closure and the normalization of $\mathcal{T}$ (Sec. 5); D9-a implements this as a band-level null. D9-b leverages the three-route equality (App. C) via $\Delta A_\star$ and flux ratios; in admissible domains (A4) both must vanish within errors.

Boundary Posture

All boundary-sensitive statements assume A4 (PT-invariant compact boundaries or standard AF/FRW fall-offs) and A5 (Nieh–Yan as boundary counterterm). Departures from these postures may alter flux diagnostics and improvement accounting. Operational anchors: covariant phase space and surface charges [13–15], and the Holst/Nieh–Yan parity bookkeeping [9–12].

Representative Anchors (One-Line Map)

Metric-affine/Einstein–Cartan: [4,6,8]. Holst/Nieh–Yan: [9–12]. Covariant phase space & boundary charges: [13–15]. CS-modified gravity: [25,26]. GW170817 speed bounds & implications: [27–34].

## Appendix G. EFT Origins of the Spurion Field

**Scope.** This appendix provides an effective–field–theory (EFT) completion that *explains* the spurion posture adopted in the main text and ties it to *projective symmetry*. At two derivatives and within A1–A5 (Sec. 2), the compensator $\epsilon(x)$ appears only through its gradient and *only* in the projectively invariant combination

$$\boxed{\mathcal{T}_\mu \equiv T_\mu - \partial_\mu \epsilon, \qquad \Gamma^\alpha{}_{\mu\nu} \to \Gamma^\alpha{}_{\mu\nu} + \delta^\alpha_\mu \xi_\nu : \ \mathcal{T}_\mu \text{ invariant.}} \tag{97}$$

We give a minimal Stueckelberg action that reproduces the C1 map, explain two equivalent implementations of the *spurion limit*, and record complementary EFT viewpoints in which the gradient-only appearance of $\epsilon$ is manifest.

*Appendix G.1. G.1 Stueckelberg completion and the spurion limit*

A projectively invariant, two-derivative completion is

$$\boxed{S_{\rm Stk} = \int d^4x \sqrt{-g}\left[\tfrac{M_P^2}{2} R(e,\omega) - \tfrac{m_T^2}{2}\mathcal{T}_\mu \mathcal{T}^\mu - \tfrac{f_\epsilon^2}{2}(\partial\epsilon)^2\right], \qquad \mathcal{T}_\mu \equiv T_\mu - \partial_\mu \epsilon,} \tag{98}$$

with $m_T > 0$ and $f_\epsilon \geq 0$ real. The masslike term for $\mathcal{T}_\mu$ *restores* projective invariance via the Stueckelberg compensator $\epsilon$; the spectator kinetic term respects the shift symmetry $\epsilon \mapsto \epsilon + c$ and keeps only $\partial\epsilon$ at this order.

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